CMAI 480

Computer Vision

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Image Alignment and Stitching

Chapter 8

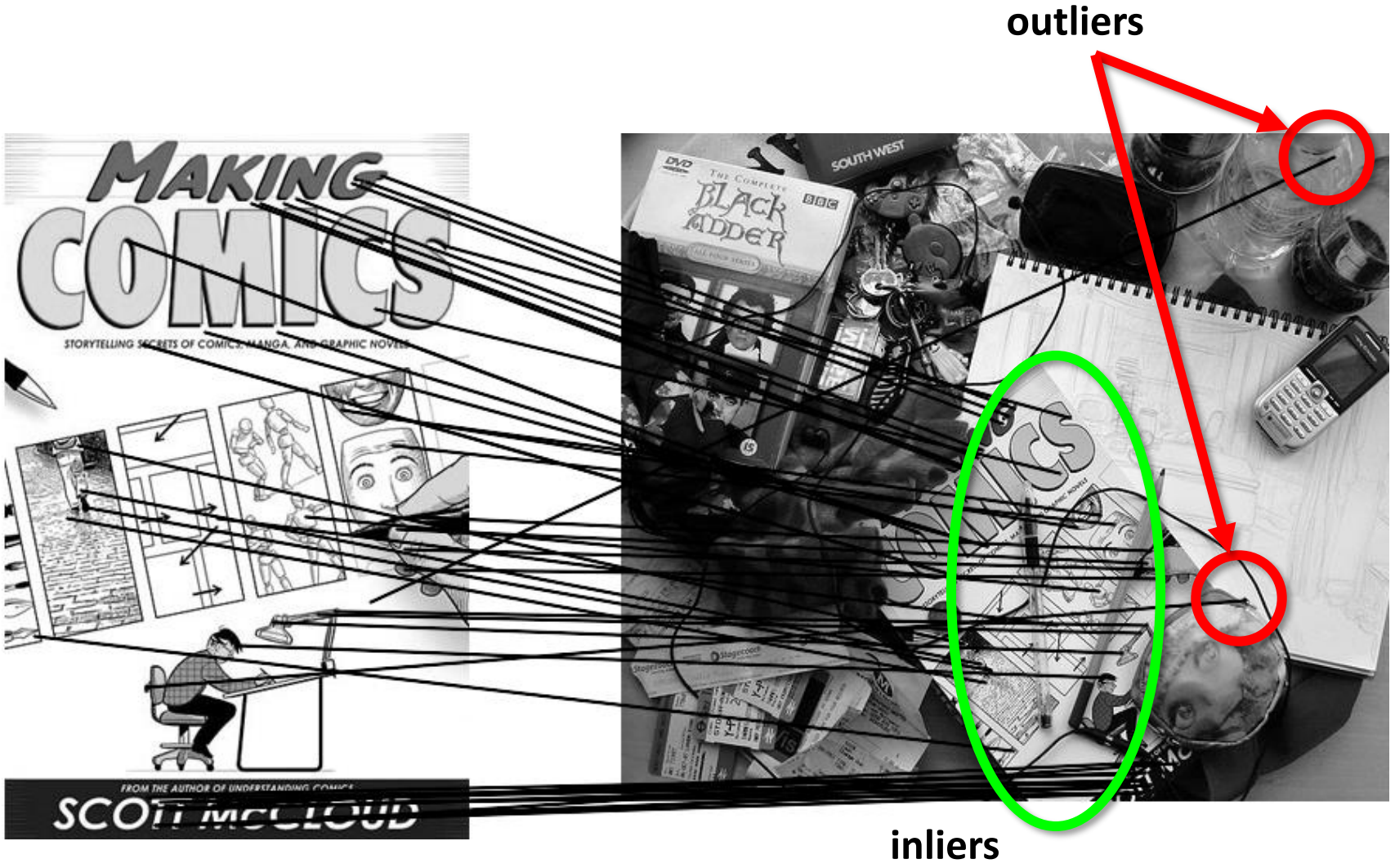
Recall: Image alignment algorithm

Given images A and B

1. Compute image features for A and B
2. Match features between A and B
3. Compute homography between A and B using least squares on set of matches

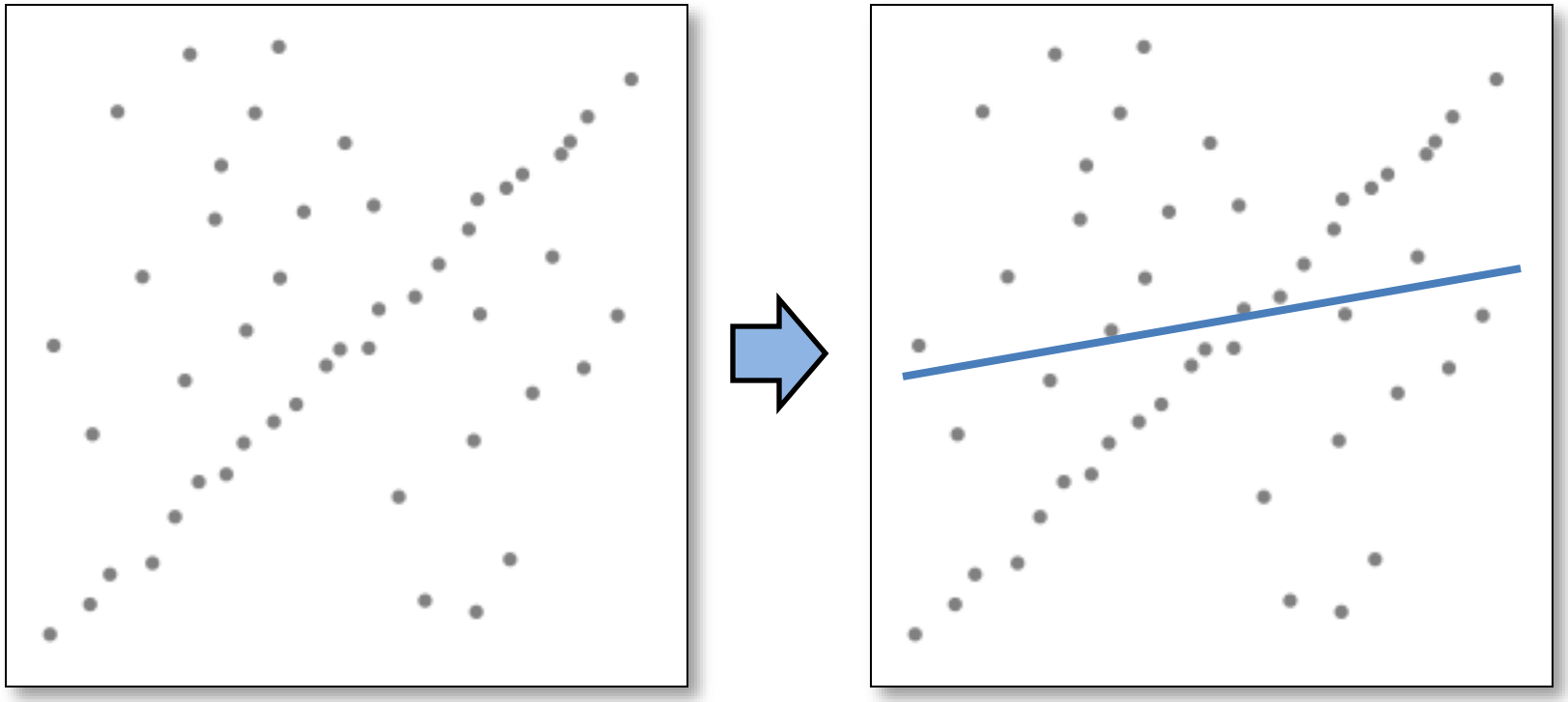
What could go wrong?

Outliers


outliers
MAKING
COMICS
STORYTELLING SECRETS OF COMICS, MANGA, AND GRAPHIC NOVELS
FROM THE AUTHOR OF UNDERSTANDING COMICS
SCOTT McCLOUD
DVD
THE COMPLETE
BLACK ADDER
BBC
SOUTH WEST
inliers

Robustness

- Let’s consider the problem of linear regression



Problem: Fit a line to these datapoints

Least squares fit

- How can we fix this?

We need a better cost function…

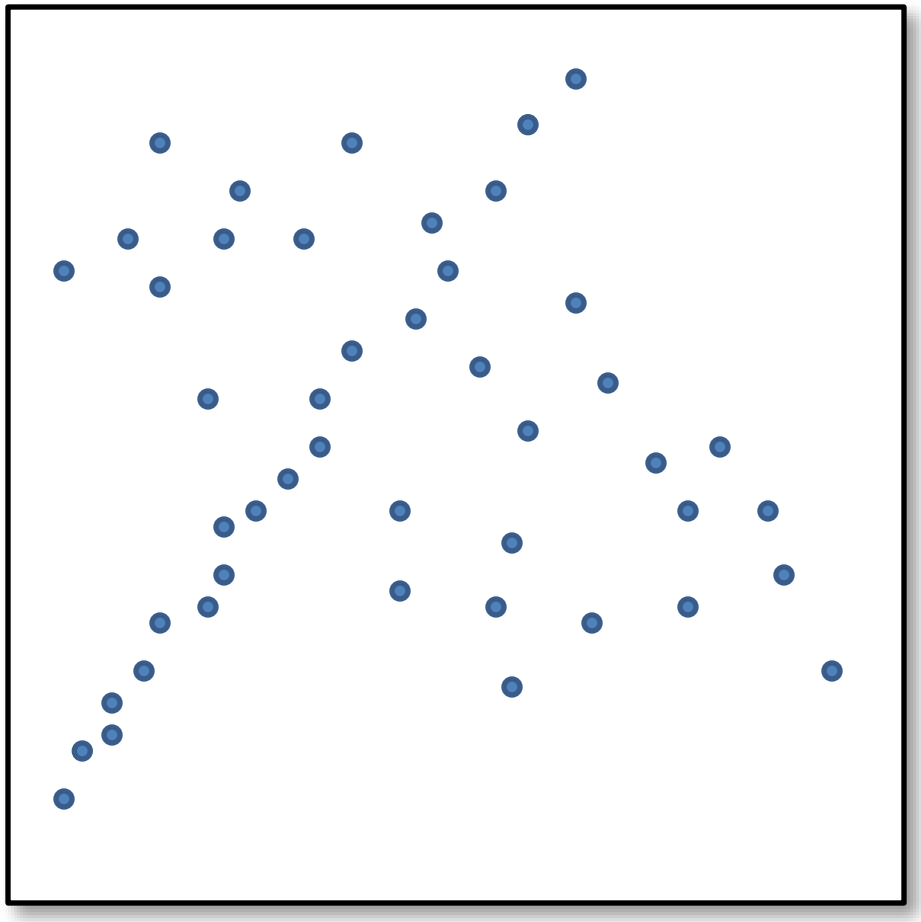
- Suggestions?

Idea

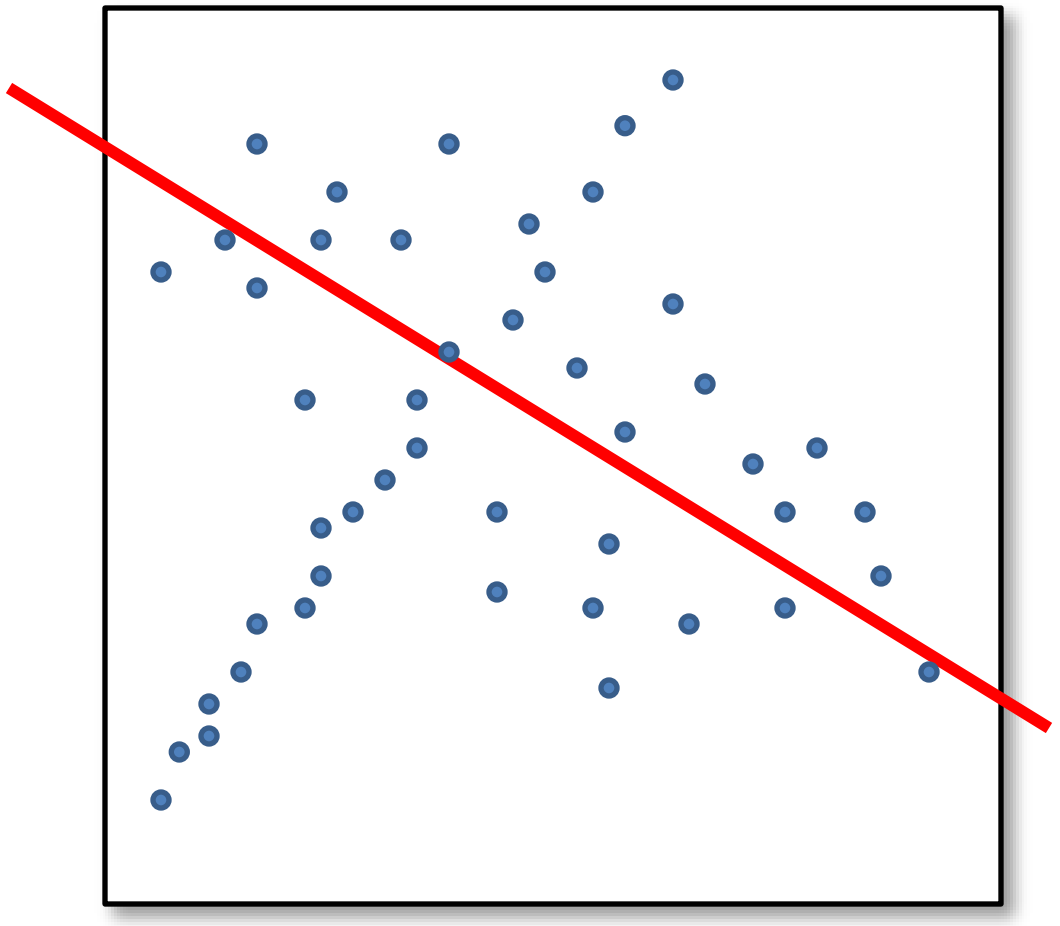
- Given a hypothesized line
- Count the number of points that “agree” with the line
  - “Agree” = within a small distance of the line
  - I.e., the **inliers** to that line

- For all possible lines, select the one with the largest number of inliers

Counting inliers

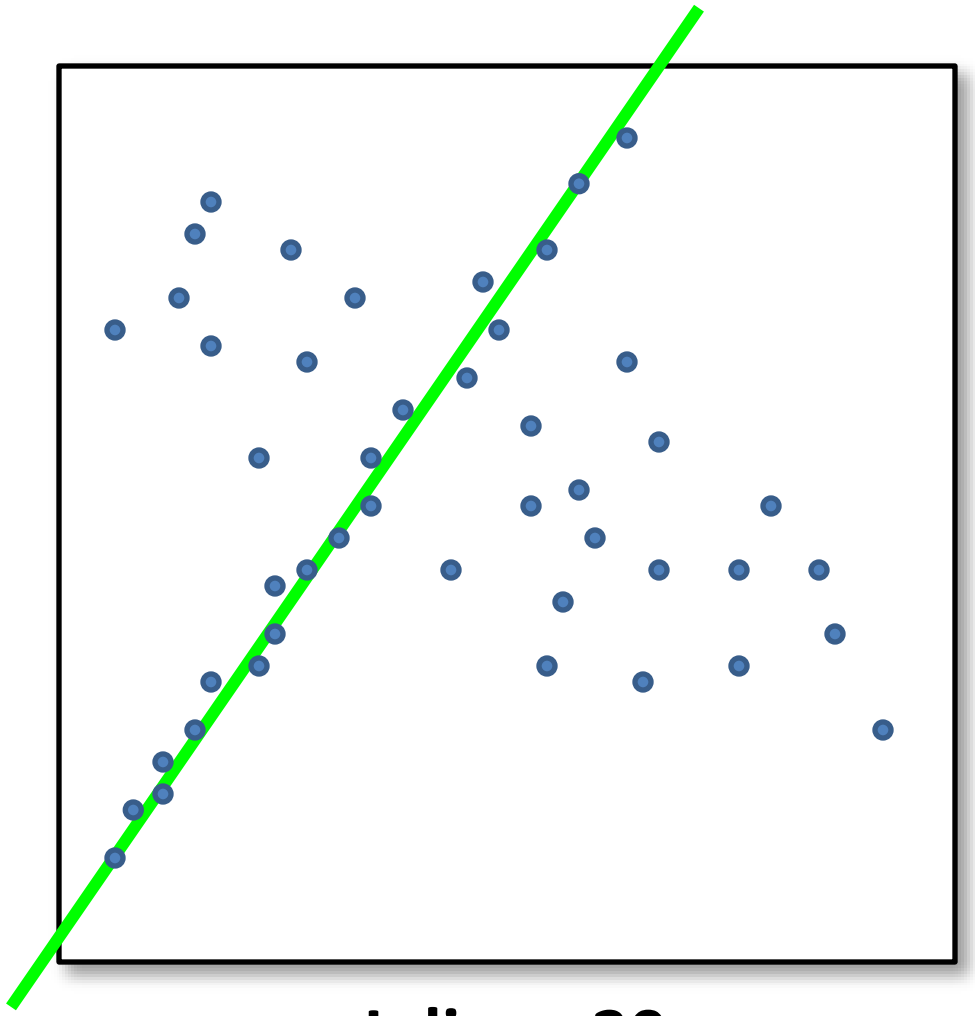


Counting inliers



Inliers: 3

Counting inliers



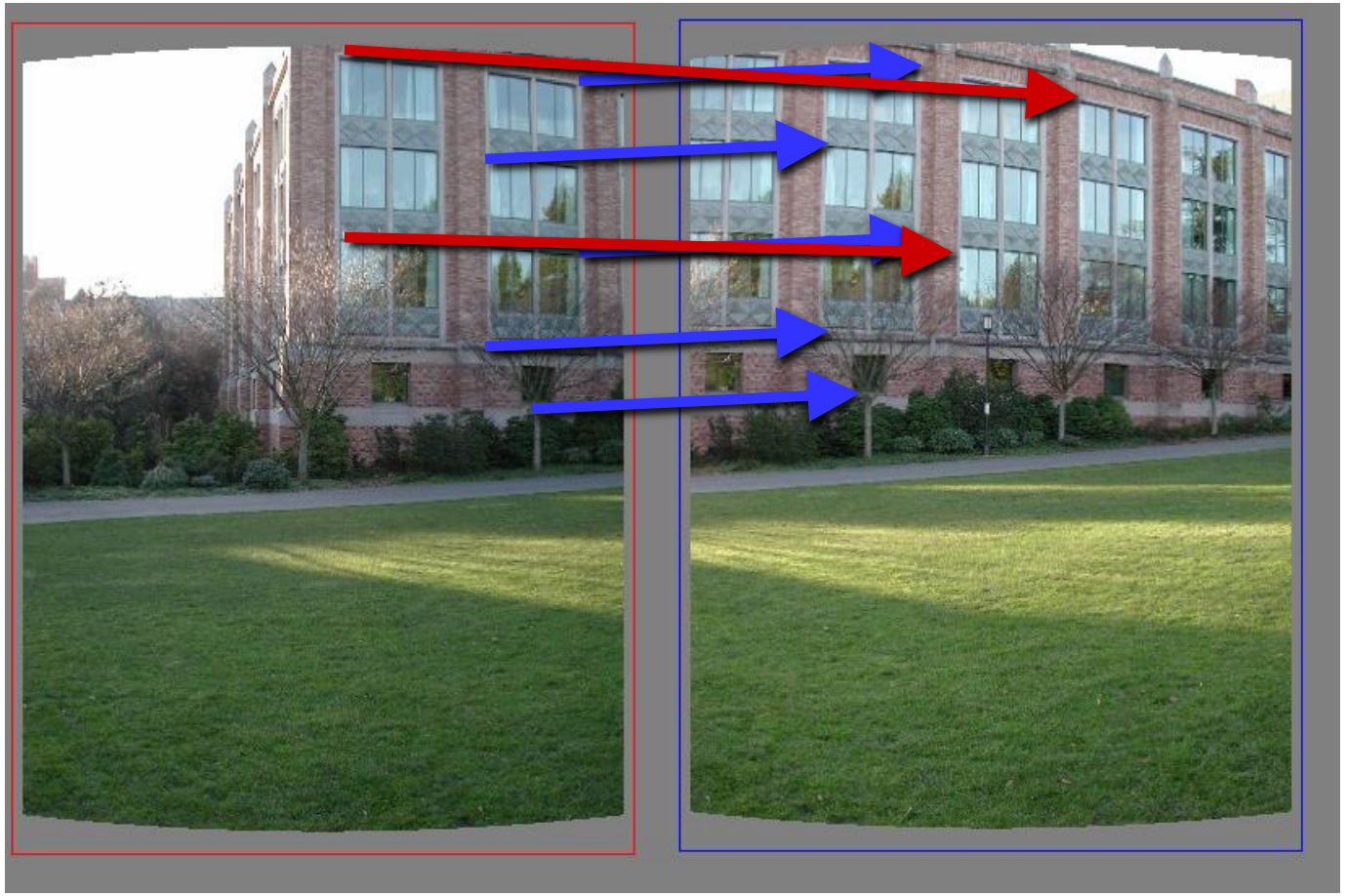
Inliers: 20

How do we find the best line?

- Unlike least-squares, no simple closed-form solution

- Hypothesize-and-test
  - Try out many lines, keep the best one
  - Which lines?

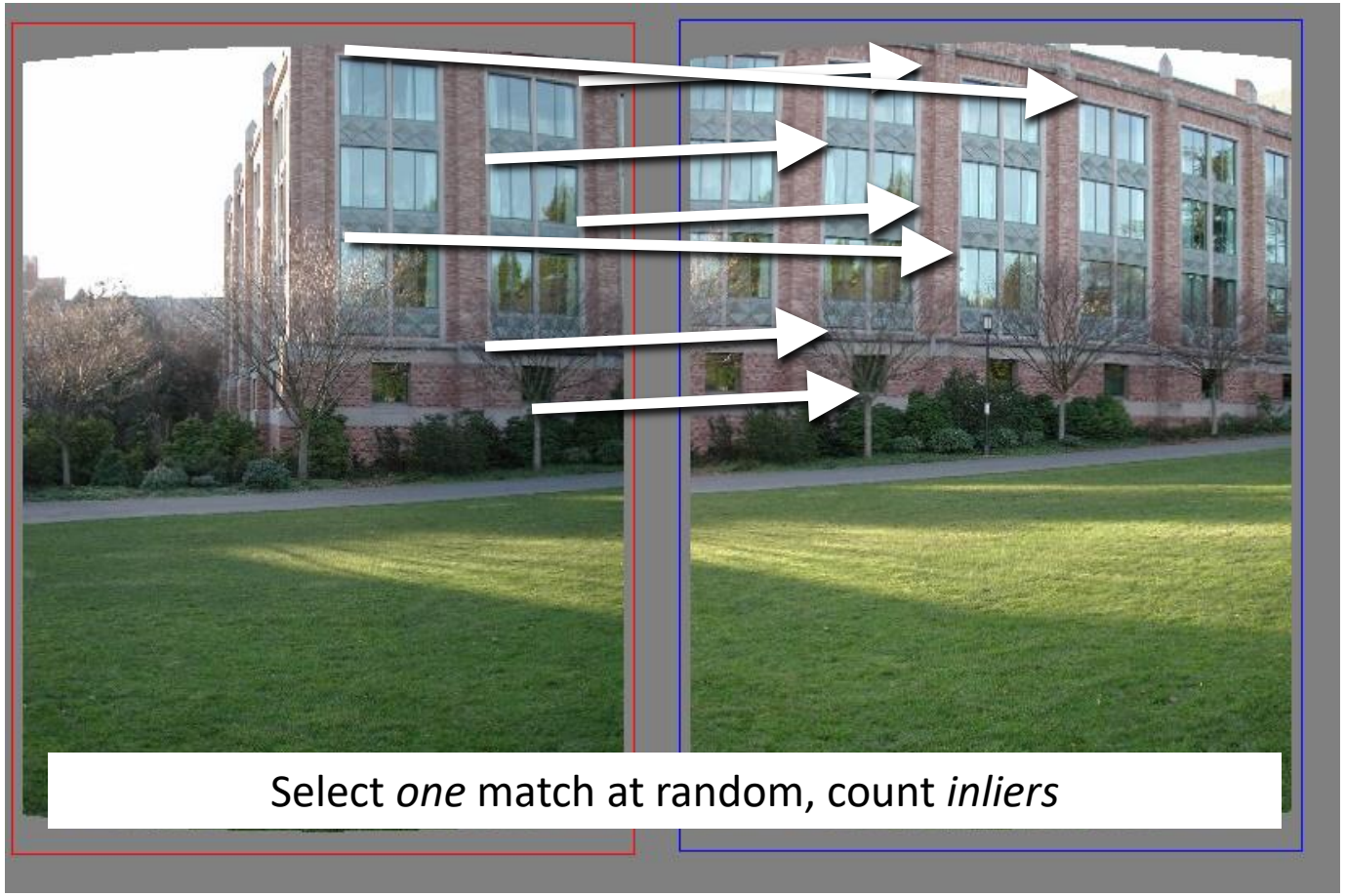
Translations



Outlier

Inlier

RAndom SAmple Consensus



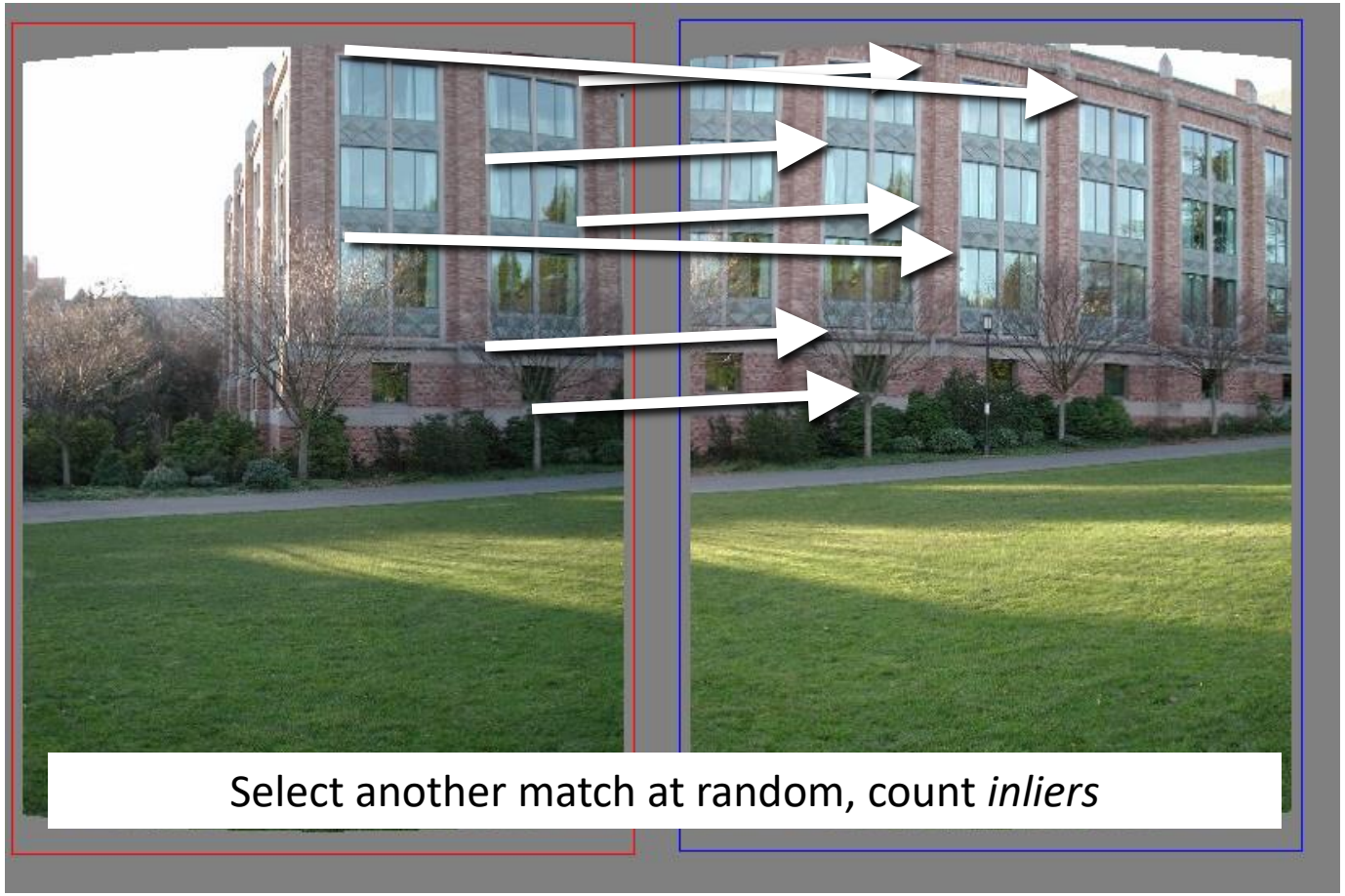
Outlier

Inlier

Match

Why one match only is needed? Remember, we are only having translation Transformation (2DoF)

RAndom SAmple Consensus

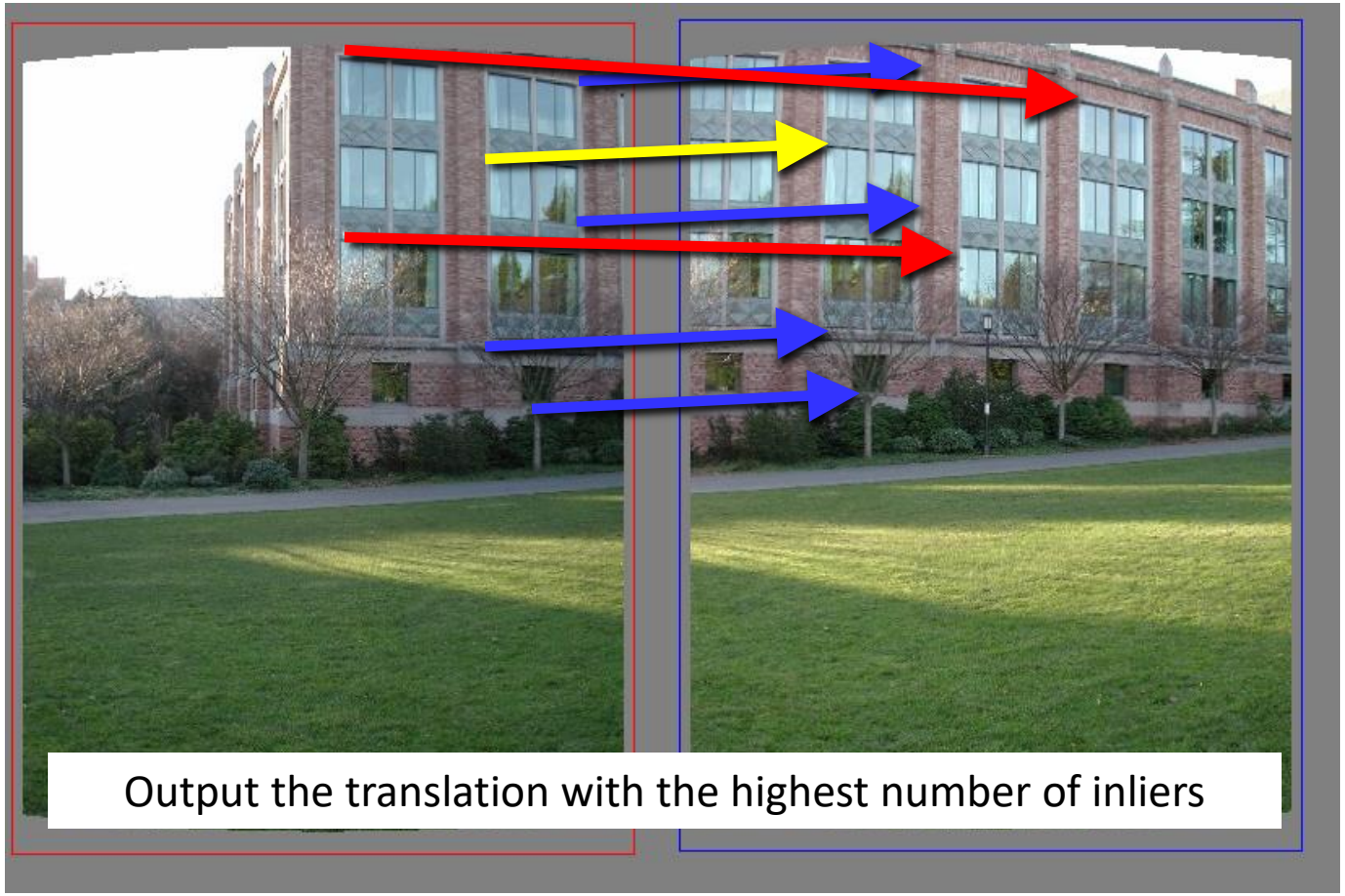


Outlier

Inlier

Match

RAndom SAmple Consensus



RANSAC: **An outlier rejection mechanism**

- Idea:
  - All the inliers will agree with each other on the translation vector; the (hopefully small) number of outliers will (hopefully) disagree with each other
    - RANSAC only has guarantees if there are < 50% outliers

Homography Estimation: RANSAC Algorithm

- Fundamental to the RANSAC algorithm is using **randomly-selected least** amount of data to construct an estimate and to then ascertain the extent of support that the rest of data provides to the estimate.
- We **accept** the estimate only if the support exceeds a threshold.
  - When an estimate made in this manner is accepted, the supporting data constitutes the **inliers** and the rest of the data the **outliers**.
- After an estimate is accepted, it can be refined using all the inliers.

RANSAC Algorithm: Parameters

- Three parameters are important to the working of the RANSAC algorithm:
  - We need a decision threshold **(δ)** to construct the inlier set.
    - When we compare a data point with the estimate, if the distance between the two is less than the threshold, we place it in the inlier set.
  - ***N*** for the number of trials to conduct.
    - It is often computationally infeasible to attempt all possible trials.
  - A minimum value **(*M*)** for the size of the inlier set for it to be acceptable.

- In most cases, the above three parameters are set by experimenting with the data.

RANSAC

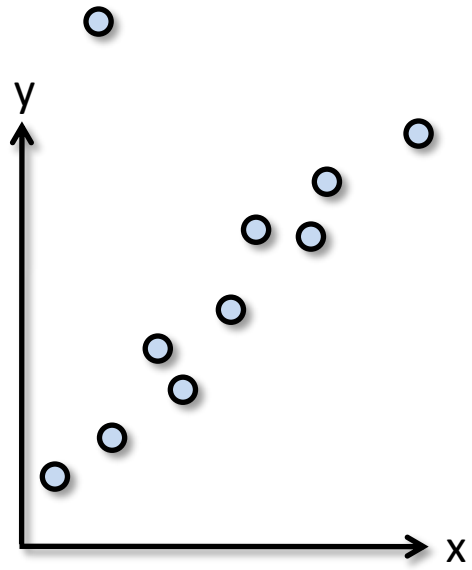
- **Inlier threshold** related to the amount of noise we expect in inliers
  - Often model noise as Gaussian w/ some standard deviation (e.g. 3 pixels)
- **Number of rounds** related to the percentage of outliers we expect, and the probability of success we'd like to guarantee
  - Suppose there are 20% outliers, and we want to find the correct answer with at least 99% probability
  - How many rounds *N* do we need?

How many rounds do we need?

$$0.2^N \leq 0.01,$$
$$N \log 0.2 \leq \log 0.01\,,$$
$$\rightarrow N \geq \frac{\log 0.01}{\log 0.2} = 3$$

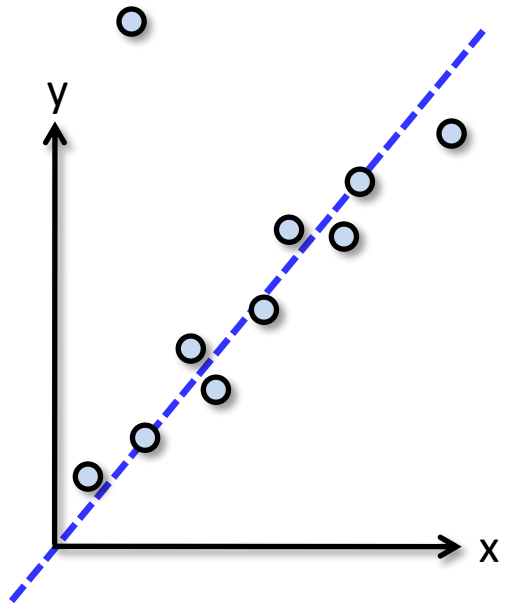
RANSAC

- Back to linear regression
- How do we generate a hypothesis?



RANSAC

- Back to linear regression
- How do we generate a hypothesis?



RANSAC

- General version:
  1. Randomly choose *s* samples
     - Typically *s* = minimum sample size that lets you fit a model
  2. Fit a model (e.g., line) to those samples
  3. Count the number of inliers that approximately fit the model
  4. Repeat *N* times
  5. Choose the model that has the largest set of inliers

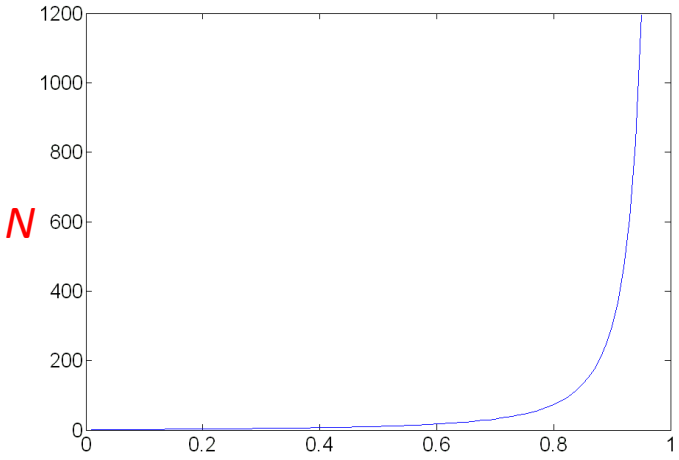
How many rounds *N*?

- If we have to choose ***s*** **samples** each time
  - with an **outlier ratio** *e*
  - and we want the **right answer with probability** *p*

$$N \geq \frac{\log(1-p)}{\log(1-(1-e)^s)}$$

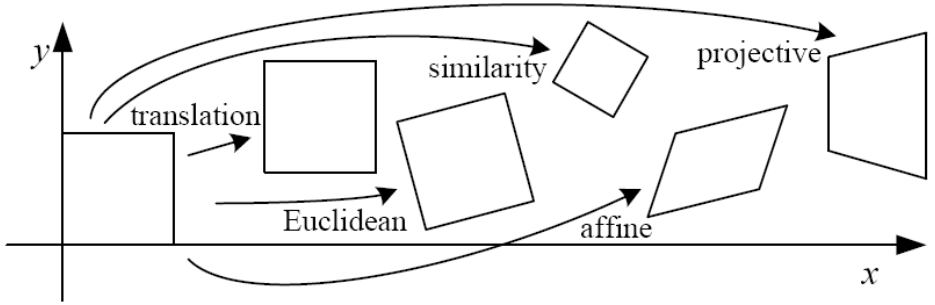
	proportion of outliers *e*						
s	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177

p = 0.99



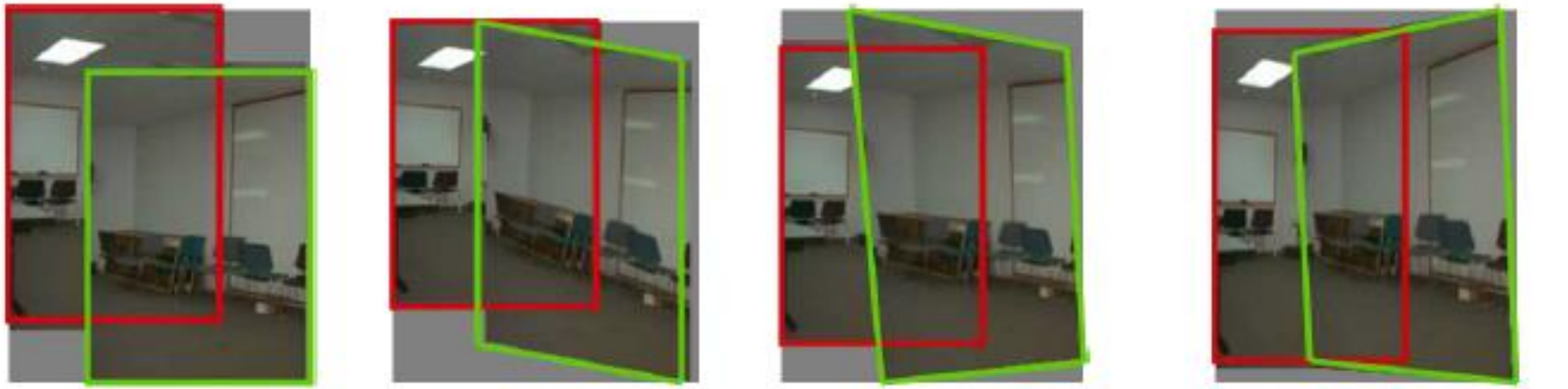
How big is *s*?

- For alignment, depends on the motion model
  - Here, each sample is a correspondence (pair of matching points)



Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\begin{bmatrix} \boldsymbol{I} \mid \boldsymbol{t} \end{bmatrix}_{2\times3}$	2	orientation $+\cdots$	
rigid (Euclidean)	$\begin{bmatrix} \boldsymbol{R} \mid \boldsymbol{t} \end{bmatrix}_{2\times3}$	3	lengths $+\cdots$	
similarity	$\begin{bmatrix} s\boldsymbol{R} \mid \boldsymbol{t} \end{bmatrix}_{2\times3}$	4	angles $+\cdots$	
affine	$\begin{bmatrix} \boldsymbol{A} \end{bmatrix}_{2\times3}$	6	parallelism $+\cdots$	
projective	$\begin{bmatrix} \tilde{\boldsymbol{H}} \end{bmatrix}_{3\times3}$	8	straight lines	

Image Stitching



(a) translation [2 dof] (b) affine [6 dof] (c) perspective [8 dof] (d) 3D rotation [3+ dof]

Figure 8.4 *Two-dimensional motion models and how they can be used for image stitching.*

RANSAC pros and cons

- Pros
  - Simple and general
  - Applicable to many different problems
  - Often works well in practice
- Cons
  - Parameters to tune
  - Sometimes too many iterations are required
  - Can fail for extremely low inlier ratios
  - We can often do better than brute-force sampling

RANSAC: Summary

- An example of a “voting”-based fitting scheme
- Each hypothesis gets voted on by each data point, best hypothesis wins

- There are many other types of voting schemes
  - E.g., Hough transforms…

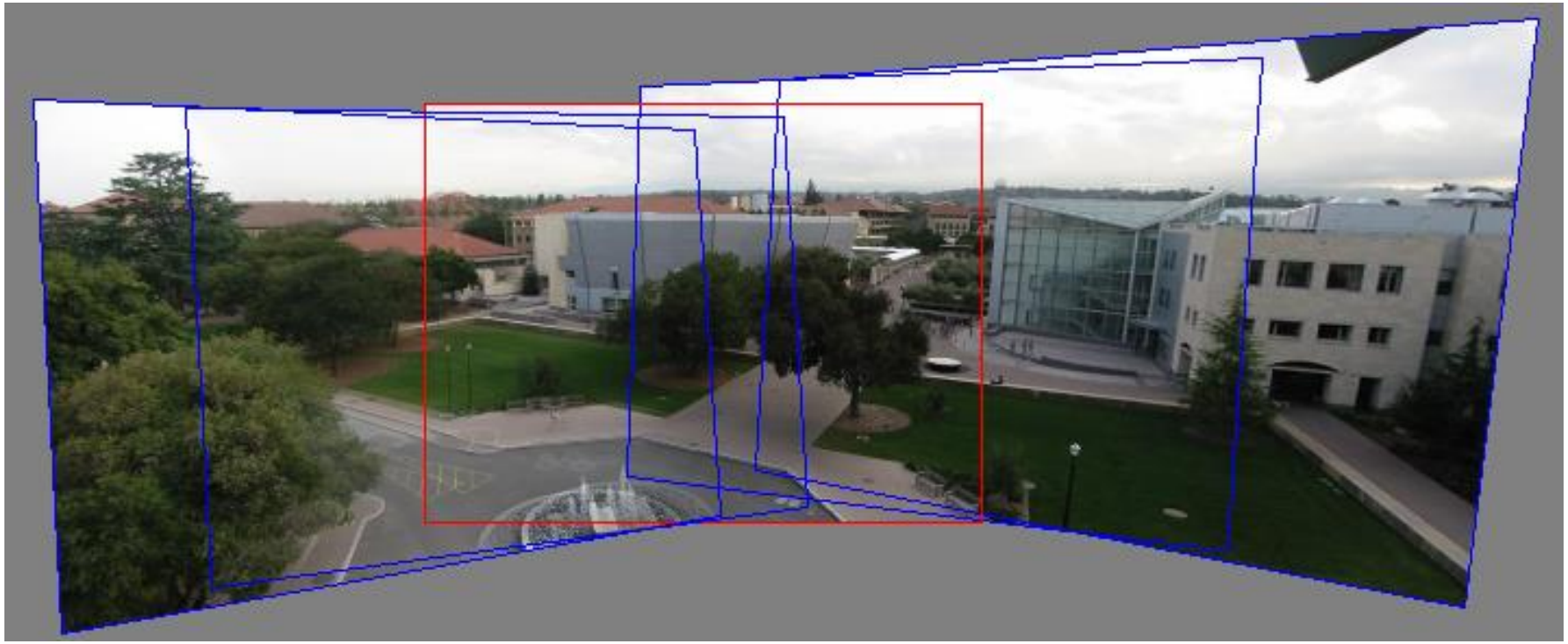
Panoramas

- Now we know how to create panoramas!

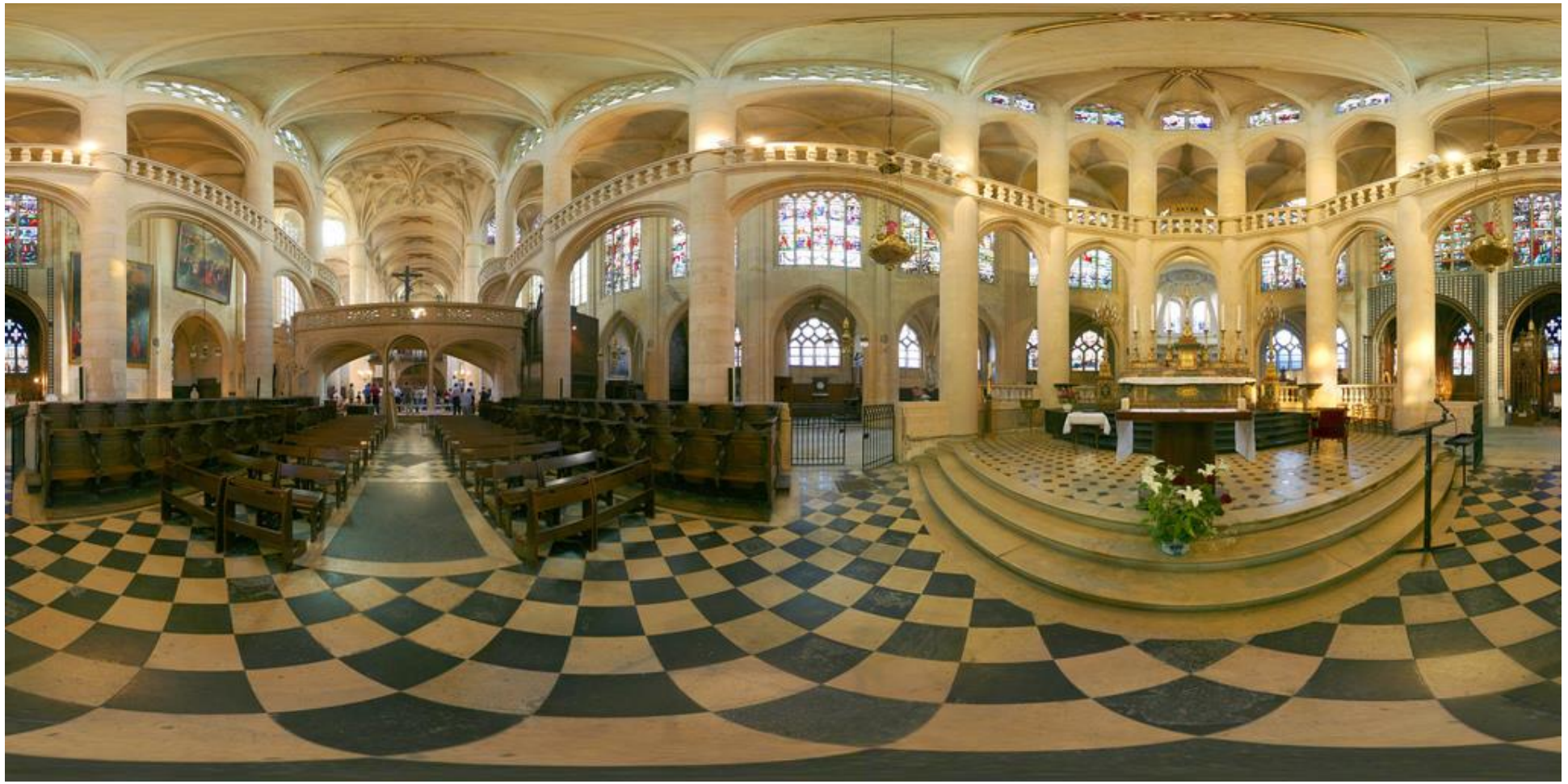
- Given two images:
  - Step 1: Detect features
  - Step 2: Match features
  - Step 3: Compute a homography using RANSAC
  - Step 4: Combine the images together (somehow)

- What if we have more than two images?

Example of a panorama



Example of a 360 panorama



How can we use homographies to create a 360 panorama?

Questions?